Modeling and Control of Cyber-Physical Systems

Final Project (Part II): Formation Control of Nonlinear Planar Mobile Robots

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I. Introduction

The objective of this final project is to govern the movement of planar mobile robots. Since the agents are heterogeneous, they exhibit different dynamic behaviours, which means they are governed by distinct system equations. The primary goal is to achieve a target configuration, specifically a triangular moving formation. To attain this, we aim to design a distributed formation controller capable of achieving and maintaining the desired formation during the robots' motion.

To tackle the complexity of the heterogeneous models, our proposed approach is structured in two main phases:

1) **Local Input-Output Feedback Linearization**: we apply a MIMO feedback linearization independently to each robot. This technique mathematically cancels the specific nonlinearities of each individual agent, transforming their distinct dynamics into a common, standardized linear model.
2) **Multi-agent system Unification and Distributed Control**: once the agents share a uniform linear structure, we aggregate their individual models to formulate the global dynamics of the linear multi-agent system. This involves defining a global state-space matrix and defining a communication graph using an adjacency and Laplacian matrix. Finally, based on this networked topology and the assigned initial conditions, the distributed formation controller is designed to drive the unified system toward the desired target configuration.

The remainder of this report is organized as follows. First, we discuss the characteristics of the nonlinear agent dynamics and the input-output feedback linearization applied to them. Then, we continue by explaining how we unified the overall model and describing the communication graph. Finally, the last sections focus on the simulation results, detailing the controller selection process and the tuning of its parameters.

II. Feedback Linearization (FL)

A. Agent structure and state variables

As we are considering planar mobile robots, their motion occurs in a 2D space. For each agent $i = 1, 2, 3$, the state vector consists of four components:

$$x_i = [p_{x,i}, v_{x,i}, p_{y,i}, v_{y,i}]^T \in \mathbb{R}^4 \tag{1}$$

representing the positions and velocities along the x and y axes.

For each of the three agents, the nonlinear model is given in the following common form:

$$\begin{cases} \dot{p}_i = v_i \\ \dot{v}_i = \phi_i(p_i, v_i) + B_i(p_i, v_i)u_i \end{cases} \tag{2}$$

The physical input applied to the system is:

$$u_i = \begin{bmatrix} u_{x,i} \\ u_{y,i} \end{bmatrix} \in \mathbb{R}^2 \tag{3}$$

Furthermore, the measured and controlled output is the planar position of the robot:

$$y_i = \begin{bmatrix} p_{x,i} \\ p_{y,i} \end{bmatrix} \in \mathbb{R}^2 \tag{4}$$

The specific dynamics of the first agent $i = 1$, which features anisotropic nonlinear drag, are represented by the following system of equations:

$$\begin{cases} \dot{p}_{x,1} = v_{x,1} \\ \dot{v}_{x,1} = -0.35v_{x,1} - 0.08v_{x,1}|v_{x,1}| + \frac{1}{1.2}u_{x,1} \\ \dot{p}_{y,1} = v_{y,1} \\ \dot{v}_{y,1} = -0.45v_{y,1} - 0.10v_{y,1}|v_{y,1}| + \frac{1}{1.2}u_{y,1} \end{cases} \tag{5}$$

The second agent $i = 2$ is characterized by a position-dependent apparent mass, and its dynamics are given by:

$$\begin{cases} \dot{p}_{x,2} = v_{x,2} \\ \dot{v}_{x,2} = -0.40v_{x,2} + \frac{1}{1+0.25\sin^2(p_{x,2})}u_{x,2} \\ \dot{p}_{y,2} = v_{y,2} \\ \dot{v}_{y,2} = -0.55v_{y,2} + \frac{1}{1+0.30\cos^2(p_{y,2})}u_{y,2} \end{cases} \tag{6}$$

Finally, the third agent $i = 3$ presents nonlinear coupling between its axes and is described by:

$$\begin{cases} \dot{p}_{x,3} = v_{x,3} \\ \dot{v}_{x,3} = -0.30v_{x,3} + 0.15\sin(p_{y,3})v_{y,3} \\ \qquad +(1 + 0.2\cos^2(p_{y,3}))u_{x,3} + 0.08\sin(p_{x,3})u_{y,3} \\ \dot{p}_{y,3} = v_{y,3} \\ \dot{v}_{y,3} = -0.35v_{y,3} + 0.15\cos(p_{x,3})v_{x,3} \\ \qquad +0.08\sin(p_{y,3})u_{x,3} + (1 + 0.2\sin^2(p_{x,3}))u_{y,3} \end{cases} \tag{7}$$

B. General Derivation of the Feedback-Linearizing Law

Notice that the models for all agents can be written in the common affine form:

$$\begin{cases} \dot{p}_i = v_i \\ \dot{v}_i = \phi_i(p_i, v_i) + B_i(p_i, v_i)u_i \end{cases} \tag{8}$$

where ϕ_i is the drift term and B_i is the decoupling matrix. Specifically:

- **Drift term** $\phi_i(p_i, v_i)$: represents the natural, unforced dynamics of the system, describes how the robot velocity evolves when no command input is applied.
- **Decoupling matrix** $B_i(p_i, v_i)$: defines how the physical control inputs directly affect the system's acceleration.

To apply the MIMO input-output feedback linearization, we must differentiate the output y_i with respect to time until the physical control input u_i explicitly appears in the equations. A fundamental theoretical requirement to satisfy is that the models must be affine in the control input. As is evident from the agents' dynamic equations, all three nodes possess nonlinear affine dynamics, easily satisfying this condition.

Taking the first time derivative of the output, we obtain:

$$\dot{y}_i = \dot{p}_i = v_i = \begin{bmatrix} v_{x,i} \\ v_{y,i} \end{bmatrix} \tag{9}$$

Since the physical input u_i does not appear in this equation, we differentiate a second time:

$$\ddot{y}_i = \dot{v}_i = \phi_i(p_i, v_i) + B_i(p_i, v_i)u_i \tag{10}$$

At this stage, the input u_i appears explicitly through the decoupling matrix B_i, hence we conclude the differentiation step. Because exactly two differentiations are required for each output component to reveal the input, the vector relative degree of the system is:

$$(r_x, r_y) = (2, 2) \tag{11}$$

Furthermore, since $r_x + r_y = 4$ and the dimension of the state is $\dim(x_i) = 4$, the input-output feedback linearization linearizes the complete dynamics, meaning there are no zero dynamics. In practical terms, this ensures that the system has no unobservable or hidden internal states that can become unstable, hence the entire physical state of the robot is fully governed by the new linearized input-output relationship.

Provided that the decoupling matrix $B_i(p_i, v_i)$ is nonsingular, we can define the feedback-linearizing control law as:

$$u_i = B_i(p_i, v_i)^{-1}[\nu_i - \phi_i(p_i, v_i)] \tag{12}$$

By substituting this control law into the second derivative equation, we cancel the nonlinearities and obtain the linearized closed-loop dynamics:

$$\ddot{y}_i = \nu_i \tag{13}$$

where $\nu_i = [\nu_{x,i}, \nu_{y,i}]^T$ acts as the new virtual control input generated by the formation controller.

C. Agent-Specific Linearization Results

To apply the theoretical framework to the three nonlinear robots, we developed a generalized algorithm. The function takes the following symbolic variables as inputs: the state vector x_i, the physical command u_i, the output y_i, and the nonlinear dynamics $\dot{x}_i$. Consequently, the algorithm automatically derives and outputs the drift term ϕ_i, the decoupling matrix B_i, and the symbolic feedback-linearizing control law u_i, which is a function of the virtual input ν_i generated by our formation controller. A pseudocode summarizing this process is provided below.

Algorithm 1 MIMO Input-Output Feedback Linearization Algorithm

procedure IO_FL_MIMO($x_i, u_i, y_i, \dot{x}_i$)
  Input: State vector x_i, physical input u_i, output y_i, system dynamics $\dot{x}_i$
  Output: Control law u_i, decoupling matrix B_i, drift term ϕ_i

  $\dot{y}_i \leftarrow J_{x_i}(y_i) \cdot \dot{x}_i$ ▷ Compute first time derivative using Jacobian
  $\ddot{y}_i \leftarrow J_{x_i}(\dot{y}_i) \cdot \dot{x}_i$ ▷ Compute second time derivative

  $B_i \leftarrow J_{u_i}(\ddot{y}_i)$ ▷ Extract decoupling matrix
  $\phi_i \leftarrow \ddot{y}_i\big|_{u_i=[0,0]^T}$ ▷ Extract drift term

  if $\det(B_i) = 0$ **then**
    return Error: Singularity detected
  end if

  $u_i \leftarrow B_i^{-1}(\nu_i - \phi_i)$ ▷ Compute feedback-linearizing control law

  return u_i, B_i, ϕ_i
end procedure

In the following subsections, the drift term ϕ_i, the decoupling matrix B_i, and the feedback-linearizing control law obtained for each individual agent are presented.

1) Agent 1: By applying Algorithm 1 to the first agent, we extracted the drift term ϕ_1 and the

decoupling matrix B_1 from its nonlinear dynamics:

$$\phi_1 = \begin{bmatrix} -0.35v_{x,1} - 0.08v_{x,1}|v_{x,1}| \\ -0.45v_{y,1} - 0.10v_{y,1}|v_{y,1}| \end{bmatrix} \tag{14}$$

$$B_1 = \begin{bmatrix} \frac{1}{1.2} & 0 \\ 0 & \frac{1}{1.2} \end{bmatrix} \tag{15}$$

Next, we verified the nonsingularity of the decoupling matrix. As expected, the determinant is constant and strictly positive:

$$\det(B_1) = \frac{1}{1.44} \approx 0.694 \neq 0 \tag{16}$$

Since the matrix is globally invertible, we successfully computed the feedback-linearizing control law u_1 as a function of the virtual input ν_1:

$$\begin{aligned} u_1 &= B_1^{-1}[\nu_1 - \phi_1] \\ &= \begin{bmatrix} 1.2(\nu_{x,1} + 0.35v_{x,1} + 0.08v_{x,1}|v_{x,1}|) \\ 1.2(\nu_{y,1} + 0.45v_{y,1} + 0.10v_{y,1}|v_{y,1}|) \end{bmatrix} \end{aligned} \tag{17}$$

2) Agent 2: Following the same procedure for the second agent, we derived the drift term ϕ_2 and the decoupling matrix B_2 from its nonlinear dynamics:

$$\phi_2 = \begin{bmatrix} -0.40v_{x,2} \\ -0.55v_{y,2} \end{bmatrix} \tag{18}$$

$$B_2 = \begin{bmatrix} \frac{1}{1+0.25\sin^2(p_{x,2})} & 0 \\ 0 & \frac{1}{1+0.30\cos^2(p_{y,2})} \end{bmatrix} \tag{19}$$

Next, we analyzed the nonsingularity of the decoupling matrix. The determinant is given by the product of its diagonal elements:

$$\det(B_2) = \frac{1}{(1 + 0.25\sin^2(p_{x,2}))(1 + 0.30\cos^2(p_{y,2}))} \tag{20}$$

Since both $\sin^2(p_{x,2}) \geq 0$ and $\cos^2(p_{y,2}) \geq 0$ for any value of the planar position, the denominator is always strictly greater than zero. Therefore, $\det(B_2) > 0$ globally. As the matrix is globally invertible, we computed the feedback-linearizing control law u_2:

$$\begin{aligned} u_2 &= B_2^{-1}[\nu_2 - \phi_2] \\ &= \begin{bmatrix} 1 + 0.25\sin^2(p_{x,2}) & 0 \\ 0 & 1 + 0.30\cos^2(p_{y,2}) \end{bmatrix} \\ &\quad \times \begin{bmatrix} \nu_{x,2} + 0.40v_{x,2} \\ \nu_{y,2} + 0.55v_{y,2} \end{bmatrix} \end{aligned} \tag{21}$$

3) Agent 3: Finally, for the third agent, which presents nonlinear coupling, the algorithm yielded the following drift term ϕ_3 and decoupling matrix B_3:

$$\phi_3 = \begin{bmatrix} -0.30v_{x,3} + 0.15\sin(p_{y,3})v_{y,3} \\ -0.35v_{y,3} + 0.15\cos(p_{x,3})v_{x,3} \end{bmatrix} \tag{22}$$

$$B_3 = \begin{bmatrix} 1 + 0.2\cos^2(p_{y,3}) & 0.08\sin(p_{x,3}) \\ 0.08\sin(p_{y,3}) & 1 + 0.2\sin^2(p_{x,3}) \end{bmatrix} \tag{23}$$

To prove that the decoupling matrix is nonsingular, we evaluate its determinant:

$$\begin{aligned} \det(B_3) = (1 + 0.2\cos^2(p_{y,3}))(1 + 0.2\sin^2(p_{x,3})) \\ - 0.0064\sin(p_{x,3})\sin(p_{y,3}) \end{aligned} \tag{24}$$

By analyzing the bounds of the trigonometric functions, the minimum possible value for the first term is $1 \times 1 = 1$, while the maximum possible value for the subtracted term is $0.08 \times 0.08 = 0.0064$. Thus, we can guarantee that globally:

$$\det(B_3) \geq 1 - 0.0064 = 0.9936 > 0 \tag{25}$$

Having proven that B_3 is strictly positive and therefore always invertible, we derive the physical input u_3 as a function of the virtual input ν_3.

$$\begin{aligned} u_3 = B_3^{-1}[\nu_3 - \phi_3] &= \frac{1}{\det(B_3)} \\ &\times \begin{bmatrix} 1 + 0.2\sin^2(p_{x,3}) & -0.08\sin(p_{x,3}) \\ -0.08\sin(p_{y,3}) & 1 + 0.2\cos^2(p_{y,3}) \end{bmatrix} \\ &\times \begin{bmatrix} \nu_{x,3} - \phi_{x,3} \\ \nu_{y,3} - \phi_{y,3} \end{bmatrix} \end{aligned} \tag{26}$$

D. The Common Double-Integrator Model

The application of the local input-output feedback linearizing law, $u_i = B_i^{-1}[\nu_i - \phi_i]$, leads to a significant structural simplification of the agents' dynamics. By design, this nonlinear feedback cancels the nonlinearities and enforces a uniform linear behavior across the entire team. For each of the three agents, the closed-loop system is

now described by the following double-integrator model:

$$\ddot{y}_i = \nu_i, \quad i = 1, 2, 3 \tag{27}$$

which can be equivalently expressed in state-space form as:

$$\begin{cases} \dot{p}_i = v_i \\ \dot{v}_i = \nu_i \end{cases} \tag{28}$$

Based on the feedback linearization process, we can draw the following conclusions:

- **Homogeneity through Normalization:** despite the agents' inherent heterogeneity, the feedback linearization acts as a normalization layer. This transformation effectively converts the diverse nonlinear dynamics of the team into a uniform set of three identical linear systems.
- **Decoupling and Linearization:** the complex nonlinear interactions of each agent are replaced by a direct, decoupled mapping between the virtual control input ν_i (representing the desired acceleration) and the system output y_i.
- **Foundation for Distributed Control:** this standardization is fundamental, as it allows us to design a single, unified formation controller that operates on the virtual input ν_i for all agents simultaneously.

III. Global Dynamics

A. Global System Matrices

The application of local MIMO input-output feedback linearization to each heterogeneous agent mathematically cancels their specific nonlinearities. Consequently, the distinct nonlinear models are transformed into a common, standardized linear double-integrator model.

For a generic agent $i \in \{1, 2, 3\}$, the state vector is defined as $x_i = [p_{x,i}, v_{x,i}, p_{y,i}, v_{y,i}]^T \in \mathbb{R}^4$. The linearized continuous-time dynamics are governed by the state-space equation:

$$\dot{x}_i = A_0 x_i + B_0 \nu_i \tag{29}$$

where $\nu_i = [\nu_{x,i}, \nu_{y,i}]^T \in \mathbb{R}^2$ represents the virtual input. The fundamental state matrix $A_0 \in \mathbb{R}^{4\times4}$ and the input matrix $B_0 \in \mathbb{R}^{4\times2}$ are structurally defined in the MATLAB implementation (`multiagents.m`) as:

$$A_0 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad B_0 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \tag{30}$$

The specific values within these matrices directly dictate the open-loop physical behavior of the linearized agent. In the A_0 matrix, the elements $A_0(1,2) = 1$ and $A_0(3,4) = 1$ encode the fundamental kinematic relationships $\dot{p}_{x,i} = v_{x,i}$ and $\dot{p}_{y,i} = v_{y,i}$. The fact that all other elements in A_0 are exactly zero implies that the linearized system possesses no inherent damping, no positional stiffness, and zero dynamic coupling between the x and y axes. The system will hold a constant velocity indefinitely if left unforced. Meanwhile, the elements $B_0(2,1) = 1$ and $B_0(4,2) = 1$ in the input matrix strictly map the virtual inputs $\nu_{x,i}$ and $\nu_{y,i}$ to the state derivatives representing accelerations ($\dot{v}_{x,i}$ and $\dot{v}_{y,i}$). Therefore, ν_i acts as a direct acceleration command.

While this structure correctly defines the isolated agents, unifying the system requires aggregating these distinct models into a single multi-agent state space. The global state vector $X \in \mathbb{R}^{12}$ is formed by concatenating the individual agent states, yielding $X = [x_1^T, x_2^T, x_3^T]^T$. Similarly, the global virtual input vector is defined as $\nu = [\nu_1^T, \nu_2^T, \nu_3^T]^T \in \mathbb{R}^6$.

To map the individual dynamics to this global state space, the global matrices $A_{global} \in \mathbb{R}^{12\times12}$ and $B_{global} \in \mathbb{R}^{12\times6}$ are constructed. In the MATLAB script, this unification is achieved using the `blkdiag` function, resulting in:

$$\begin{aligned} A_{global} &= \text{blkdiag}(A_0, A_0, A_0) \\ &= \begin{bmatrix} A_0 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & A_0 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & A_0 \end{bmatrix} \end{aligned} \tag{31}$$

$$\begin{aligned} B_{global} &= \text{blkdiag}(B_0, B_0, B_0) \\ &= \begin{bmatrix} B_0 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & B_0 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & B_0 \end{bmatrix} \end{aligned} \tag{32}$$

The block-diagonal nature of these global matrices is physically highly significant. Because all off-diagonal block matrices are strictly zero ($\mathbf{0} \in \mathbb{R}^{4\times4}$), it mathematically guarantees that the state of one agent has absolutely no natural influence on the state of another. The unified equation $\dot{X} = A_{global}X + B_{global}\nu$ therefore represents a system of three decoupled, freely moving bodies. This formulation establishes the required baseline for a multi-agent framework: since there is no inherent physical or dynamic connection between

the robots embedded in A_{global}, any coordinated behavior or formation tracking must be entirely engineered by the higher-level controller. This necessitates the definition of a specific communication topology to actively share state information between the independent models.

B. Communication Graph Topology

The global state-space representation defined by A_{global} models the multi-agent system as three completely uncoupled entities. Because the agents possess no inherent physical connections, achieving a coordinated formation requires them to actively share state information. This networked communication is mathematically modeled using algebraic graph theory, where the mobile agents are represented as nodes and the active data-sharing links are represented as edges.

In this project, the communication topology is designed as a directed chain, mathematically enforcing a strict hierarchical flow of information denoted as $1 \rightarrow 2 \rightarrow 3$. Within this hierarchy, Agent 1 is designated as the formation leader. Agent 2 acts as a follower that receives state information exclusively from Agent 1, while Agent 3 is a subsequent follower that receives information exclusively from Agent 2.

This specific topology is encoded in the MATLAB implementation (`multiagents.m`) using the adjacency matrix $\mathcal{A} \in \mathbb{R}^{3\times3}$. The elements of this matrix are defined such that $\mathcal{A}_{ij} > 0$ if node i receives information from node j, and $\mathcal{A}_{ij} = 0$ otherwise. For the chosen directed chain, the adjacency matrix takes the form of a lower shift matrix:

$$\mathcal{A} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \tag{33}$$

To quantify the localized communication required for distributed control, the network is further characterized by its in-degree Laplacian matrix, $\mathcal{L} \in \mathbb{R}^{3\times3}$. The Laplacian is computed as $\mathcal{L} = \mathcal{D} - \mathcal{A}$, where $\mathcal{D}$ is the diagonal in-degree matrix containing the row sums of $\mathcal{A}$. The Laplacian physically represents the relative state discrepancies between neighbor agents, functioning algebraically like a network of virtual springs connecting the nodes. For the 1-2-3 chain topology, the computed Laplacian is:

$$\mathcal{L} = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \tag{34}$$

A fundamental property of $\mathcal{L}$ for achieving consensus is the presence of a spanning tree, which guarantees that information from a root node can reach all other nodes in the network. In the MATLAB script, this connectivity is computationally verified by analyzing the eigenvalues of the Laplacian (`L_chain`); the presence of exactly one zero eigenvalue confirms that the directed graph is connected and rooted at Agent 1.

Furthermore, the overall control objective requires the entire formation to track a dynamic moving reference trajectory in the global plane. Under the distributed paradigm, it is assumed that not all agents have access to this global reference. Instead, a pinning control strategy is implemented. The reference trajectory data is exclusively injected into the leader (Agent 1). This is mathematically structured using a pinning matrix $P_{in} = \text{diag}(1, 0, 0)$, scaled by a scalar pinning gain $\gamma = 1.2$.

The complete stability and convergence properties of this tracking network are completely governed by the pinned Laplacian matrix, $\mathcal{L}_p$, which merges the relative inter-agent communication ($\mathcal{L}$) with the absolute reference tracking (γP_{in}):

$$\mathcal{L}_p = \mathcal{L} + \gamma P_{in} = \begin{bmatrix} 1.2 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \tag{35}$$

Because $\mathcal{L}_p$ is a lower triangular matrix, its eigenvalues ρ_i are explicitly defined by its diagonal elements, yielding the spectrum $\rho = \{1.2, 1, 1\}$. The fact that all eigenvalues possess strictly positive real parts guarantees that the networked system is structurally stable; the reference information injected at the leader will successfully permeate through the entire chain without diverging. Crucially, the minimum eigenvalue, $\rho_{min} = 1$, quantifies the algebraic connectivity of the pinned network. Physically, ρ_{min} defines the algebraic connectivity of the pinned network, which dictates the theoretical lower bound on the convergence rate of the consensus protocol, effectively acting as the mathematical bottleneck for information propagation across the graph. Identifying this precise value is a critical analytical step, as it serves as the foundational parameter upon which the subsequent distributed formation controller must be tuned to guarantee overall system stability.

C. Target Formation

With the multi-agent system unified and the communication topology established, the final structural requirement is to mathematically define the target configuration. The objective is to coordinate the robots into a specific moving geometric shape. Rather than prescribing absolute global coordinates for every agent, the formation is defined relatively, leveraging the communication graph.

The target configuration for this project is a rigid, equilateral triangular formation with a side length of $\ell = 2$ meters. The leader (Agent 1) is positioned at the foremost tip of the triangle, while Agents 2 and 3 constitute the left and right vertices of the trailing base, respectively. To integrate these geometric constraints into the multi-agent state-space framework, a formation offset matrix $H \in \mathbb{R}^{4\times 3}$ is constructed. Each column H_i defines the nominal spatial offset for a respective agent in the full four-dimensional state space. Because the desired formation dictates a rigid structure with zero relative velocity between agents, the velocity components within these offset vectors are strictly zero. Based on the MATLAB implementation, the spatial arrangement relative to the leader's local frame is defined as:

$$\begin{aligned} H &= \begin{bmatrix} H_1 & H_2 & H_3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -\ell\frac{\sqrt{3}}{2} & -\ell\frac{\sqrt{3}}{2} \\ 0 & 0 & 0 \\ 0 & -\frac{\ell}{2} & \frac{\ell}{2} \\ 0 & 0 & 0 \end{bmatrix} \end{aligned} \tag{36}$$

To maintain this spatial formation while navigating the global plane, a circular reference trajectory is generated. The reference state $x_r \in \mathbb{R}^4$ provides the desired global positions and velocities, while a feedforward acceleration term $u_{ff} \in \mathbb{R}^2$ is computed to optimize dynamic tracking. As established by the pinning control strategy, this absolute reference is only available to the leader.

For the follower agents to acquire and maintain the formation, they must evaluate their position relative to their neighbors. This is achieved by computing a localized, distributed state error ϵ_i. For each agent i, the error evaluates the discrepancies between its own state x_i and the state of its transmitting neighbor x_j, explicitly compensated by their respective geometric offsets from the matrix H:

$$\epsilon_i = \sum_{j=1}^{3} \mathcal{A}_{ij} \left[(x_j - x_i) - (H_j - H_i) \right] \tag{37}$$

This error formulation perfectly bridges the geometric formation matrix H with the communication adjacency matrix $\mathcal{A}$. It effectively computes the required state correction relying exclusively on authorized communication links. This distributed error vector ϵ_i acts as the fundamental input signal for the higher-level formation controller, providing the necessary mathematical framework to synthesize the virtual acceleration commands ν_i.

Furthermore, the versatility of this architecture allows for the simulation of the multi-agent system across different spatial formations. Because the geometric configuration is entirely abstracted within the formation offset matrix H and the distributed error ϵ_i, the physical shape of the drone fleet can be seamlessly modified without fundamentally altering the network topology or the underlying control laws. Alternative target formations can be evaluated by simply redefining the spatial offset vectors H_i, allowing for a direct analysis of how different geometric constraints affect the transient maneuvering of the agents and the physical effort required to acquire the target shape.

IV. State Feedback Controller

A. Theoretical Framework for Distributed Formation Control

A distributed state-feedback controller can be employed to drive the agents toward a desired formation. To achieve this, each node relies on relative state information shared through the communication network to minimize its neighborhood formation error ϵ_i. The fundamental local control law for the generic agent is defined as:

$$\nu_i = cK\epsilon_i \tag{38}$$

where $c > 0$ represents the scalar coupling gain and K is the feedback gain matrix.

Furthermore, to navigate the formation through space, this architecture can be extended by assigning a single node to act as a sort of leader. This leader is responsible for tracking an external reference trajectory, while the remaining nodes maintain the formation shape by following the

leader's motion through the network interactions. After selecting a reference node (typically Node 1), the local formation error δ_i for each of the remaining agents can be computed. This variable measures the discrepancy between the actual relative state of the agents and the desired relative geometry:

$$\delta_i = (x_i - x_1) - (h_i - h_1), \quad i = 2, 3, \ldots, N \tag{39}$$

where x_i is the state of the i-th node and h_i represents its target coordinates within the desired static formation.

By collecting the local errors of all the follower nodes, we can define the global relative formation error vector δ and the desired relative formation vector δ_H as follows:

$$\delta = \text{col}(\delta_2, \delta_3, \ldots, \delta_N) \tag{40}$$

$$\delta_H = \text{col}(h_2 - h_1, h_3 - h_1, \ldots, h_N - h_1) \tag{41}$$

Using these aggregated definitions, the global formation error dynamics for the multi-agent system can be expressed as:

$$\dot{\delta}(t) = A_c \delta(t) + (I_{N-1} \otimes A)\delta_H \tag{42}$$

where A_c is the closed-loop formation state matrix, given by:

$$\begin{aligned} A_c = {} & I_{N-1} \otimes A \\ & - (L_{22} + 1_{N-1}\alpha^T) \otimes CBK \end{aligned} \tag{43}$$

In this expression, L_{22} and α^T represent the sub-blocks of the partitioned Laplacian matrix obtained by isolating the reference Node 1.

Considering the previously defined control law, the formation control problem is solvable if and only if the following three conditions are strictly satisfied:

- The communication graph contains a directed spanning tree.
- The coupling gain c and the feedback gain matrix K are chosen such that the closed-loop state matrix A_c is Hurwitz.
- The desired relative formation vector δ_H is selected to satisfy the following geometric constraint:

$$(I_{N-1} \otimes A)\delta_H = 0 \tag{44}$$

To achieve a moving formation, the defined control protocol must be extended to allow the entire multi-agent system to track a desired trajectory. This is accomplished by introducing additional control terms. Recalling that for our feedback-linearized agents the distributed formation controller acts on the virtual input ν_i, the control law for the follower agents $i = 2, 3, \ldots, N$ becomes:

$$\nu_i = cK\epsilon_i + \nu_{ff}(t) \tag{45}$$

Conversely, an additional local tracking loop is applied to the leader to anchor the entire formation to the prescribed reference trajectory $x_r(t)$:

$$\nu_1 = cK\epsilon_1 + \nu_{ff}(t) + \nu_{fb}(t) \tag{46}$$

The components of these control laws serve distinct physical and control purposes:

- **Relative formation term ($cK\epsilon_i$):** This distributed proportional term relies on the neighborhood state information to enforce and preserve the desired geometric shape of the formation.
- **Feedforward term ($\nu_{ff}(t)$):** This term generates the collective translational motion of the team. It represents the nominal input required by the single-agent model to execute the reference trajectory $x_r(t)$.
- **Local tracking feedback ($\nu_{fb}(t)$):** Applied exclusively to the leader, this closed-loop correction is defined as $\nu_{fb}(t) = K_1(x_r(t) - x_1(t))$, where K_1 is the local tracking gain. It forces the leader to strictly track the external reference, effectively pulling the rest of the networked formation along the desired path.

To compute the stabilizing feedback gain matrix K, an optimal control approach based on the Algebraic Riccati Equation (ARE) is employed. Given two symmetric positive-definite weighting matrices Q and R, we solve for the unique positive-definite matrix P in the continuous-time ARE:

$$A^T P + PA - PBR^{-1}B^T P + Q = 0 \tag{47}$$

The feedback gain matrix is then obtained as:

$$K = R^{-1}B^T P \tag{48}$$

Finally, the leader's local tracking gain is chosen proportionally to K, such that $K_1 = c_1 K$, where c_1 regulates the tracking strength.

In this framework, Q and R act as fundamental tuning parameters to balance performance and energy consumption:

- **Matrix Q (State Penalty):** Penalizes tracking errors. Larger values force an aggressive response to deviations, leading to a fixed formation at the cost of higher physical effort.
- **Matrix R (Control Penalty):** Penalizes the control effort (the virtual acceleration ν_i). Larger values prioritize energy savings, but resulting in slower convergence to the desired shape.

B. Experiments and Results

This section presents the numerical simulations conducted to validate the proposed multi-robot formation control laws. The analysis follows an incremental approach: initially, the system is evaluated under nominal ideal conditions, characterized by no disturbances and perfect sensor measurements, to verify the asymptotic convergence of the state feedback controller across different communication topologies.

1) Nominal Scenario (Ideal Conditions): In this baseline scenario, the state variables of the three robots (R_1, R_2, R_3) are assumed to be instantaneously and precisely measurable. The control objective requires the agents to achieve and maintain a rigid triangular geometry while the leader agent (R_1) tracks a circular reference trajectory defined by center $(5, 5)$ and radius $r = 3$ m.

To evaluate the impact of network connectivity, the same state feedback law is tested under two distinct directed communication topologies: Graph 1 (chain topology, $1 \to 2 \to 3$) and Graph 2 (bidirectional follower link, $1 \to 2 \leftrightarrow 3$). Across both topologies, the agents depart from identical initial conditions: $R_1(8, 5)$, $R_2(2, -1)$ and $R_3(-1, 3)$. The leader R_1 instantly acquires and traverses the circular reference at a constant angular velocity.

Graph 1 (Figure 1): The followers R_2 and R_3 execute a smooth transient maneuver. Due to the strictly unidirectional flow of information, R_3 updates its trajectory solely based on the states of R_2, resulting in a wider, delayed spatial realignment loop.

Graph 2 (Figure 2): The introduction of the bidirectional communication link between the followers $(2 \leftrightarrow 3)$ significantly modifies the transient kinematics. The mutual state exchange forces R_2 and R_3 into tighter, highly coupled spatial maneuvers.

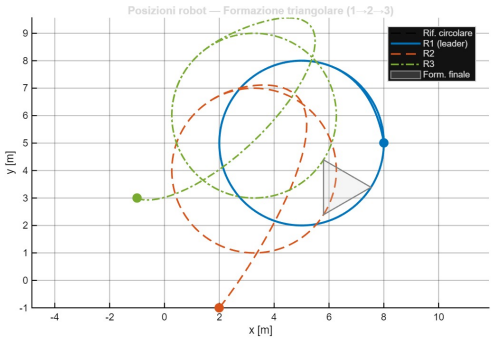


Figure 1: Trajectories of the agents under Graph 1 (chain topology) in nominal conditions.

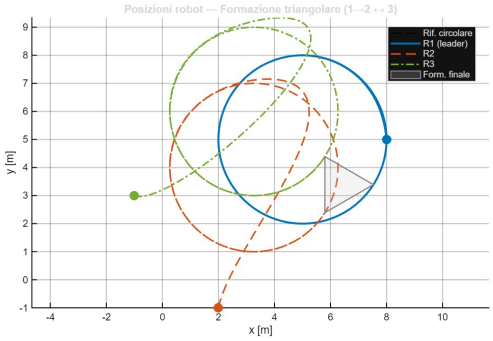


Figure 2: Trajectories of the agents under Graph 2 (bidirectional follower link) in nominal conditions.

Across both tested topologies, the tracking error $\|e_{\text{track}}\|$ and the formation errors $\|e_{\text{form}}\|$ converge exponentially to zero:

$$\lim_{t\to\infty} \|e(t)\| = 0 \tag{49}$$

This steady-state result is guaranteed because both graphs contain a directed spanning tree rooted at the leader agent (R_1). Under this topological condition, the Laplacian matrix $\mathcal{L}$ associated with the network possesses a single zero eigenvalue associated with the consensus eigenvector, while all other eigenvalues have strictly positive real parts. Since the state feedback law places the closed-loop poles strictly in the open left half-plane $(\text{Re}(\lambda_i) < 0)$, the autonomous error dynamics are globally asymptotically stable.

The structural divergence in transient overshoot observed between Graph 1 (Figure 3) and Graph 2 (Figure 4) is dictated by the algebraic connectivity of the network, quantified by the second smallest eigenvalue of the Laplacian, $\lambda_2(\mathcal{L})$:

- **In Graph 1 (Unidirectional Chain):** The Laplacian matrix is strictly lower-triangular.

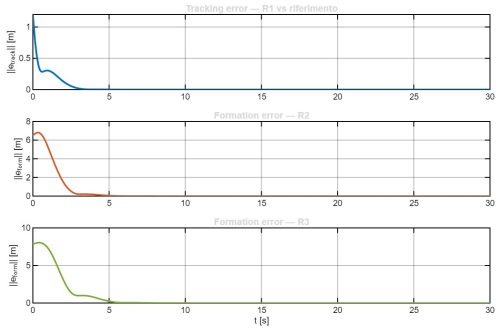


Figure 3: Tracking and formation errors for Graph 1.

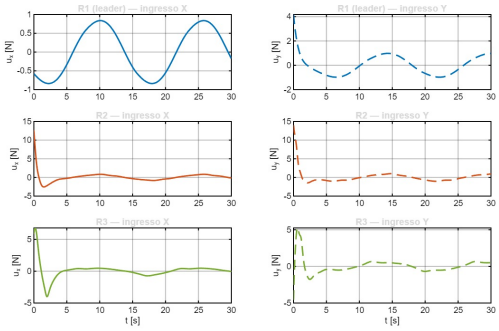


Figure 5: Control input force profiles $u_x(t)$ and $u_y(t)$ for Graph 1.

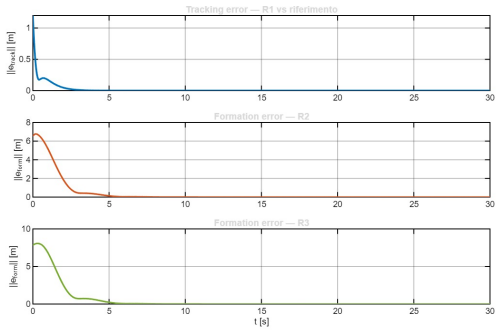


Figure 4: Tracking and formation errors for Graph 2.

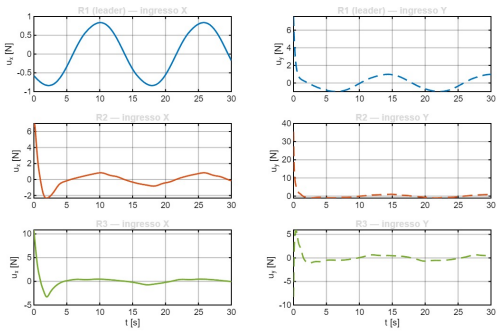


Figure 6: Control input force profiles $u_x(t)$ and $u_y(t)$ for Graph 2.

Consequently, the system operates as a dynamic cascade: agent R_3 receives zero direct information regarding the leader and updates its state solely based on R_2. This structural delay forces R_3 to accumulate the tracking error of R_2, generating a phase lag that justifies the higher transient formation error peak.

- **In Graph 2 (Bidirectional Coupling):** The mutual communication link ($2 \leftrightarrow 3$) introduces symmetric off-diagonal terms in $\mathcal{L}$, increasing the algebraic connectivity $\lambda_2(\mathcal{L})$. When R_2 accelerates to close its gap with the leader, the state error of R_3 instantly exerts a mathematical damping effect on R_2's motion. This cooperative redistribution of the trajectory error suppresses the transient peak.

The input force profiles $u_x(t)$, $u_y(t)$ (Figure 5 and Figure 6) reflect a sharp dynamic transition between two distinct operational regimes:

1) **The Proportional Transient:** the control law $u = -Kx$ maps the large initial geometric dislocation directly into an impulsive control effort. In Graph 1, agent R_2 absorbs this initial corrective demand entirely individually. In Graph 2, the mutual state feedback shares this initial kinematic burden between the two followers.
2) **The Dynamic Feedforward Regime:** Once the agents lock into the rigid triangular geometry, the linear feedback term naturally vanishes. The residual low-amplitude harmonic oscillations observed in the inputs represent the exact centripetal force required by the uncoupled agent models to sustain uniform circular motion.

To conclude, the nominal simulations demonstrate that while state feedback is theoretically sufficient to drive the formation error to zero regardless of the link orientation, the graph topology acts as a spatial filter on the transient dynamics. Higher edge density does not alter the final equilibrium, but optimizes the transient energy distribution and accelerates the rate of convergence.

2) Robustness Analysis: Sensor Noise: To evaluate the disturbance rejection capabilities of the Riccati-optimized state feedback controller (K), stochastic noise is injected into the Cartesian

control channels u_x, u_y of all agents.

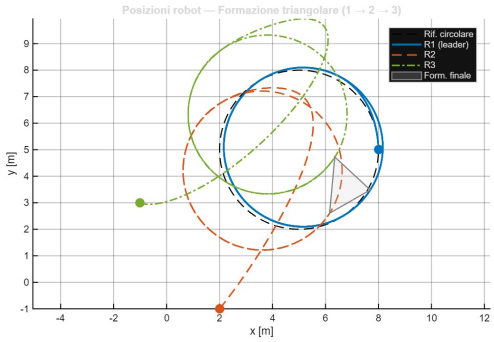


Figure 7: Agent trajectories under stochastic sensor noise.

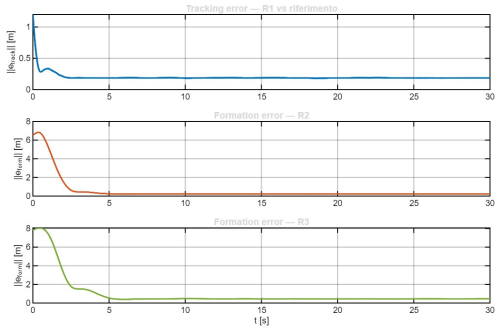


Figure 8: Tracking and formation errors under stochastic sensor noise.

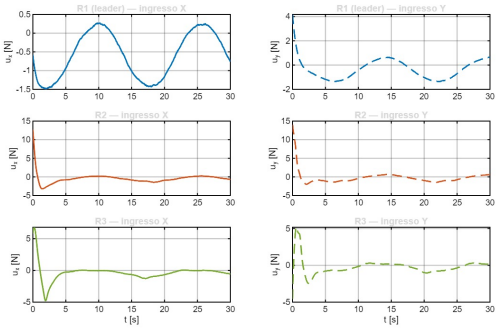


Figure 9: Control input force profiles under stochastic sensor noise.

Under these perturbed conditions (Figure 7), the steady-state tracking error of the leader shifts to 0.0118 m, while the formation errors settle at 0.0185 m (R_2) and 0.0150 m (R_3).

3) Robustness Analysis: Combined Noise and External Disturbances: To push the multi-agent architecture to its analytical limits, the Riccati-optimized state feedback law (K) is subjected to a combined stress test: stochastic input noise is maintained, while persistent external disturbances $d(t)$ are injected into the operational environment. Two distinct adversarial regimes are evaluated on the chain topology ($1 \rightarrow 2 \rightarrow 3$).

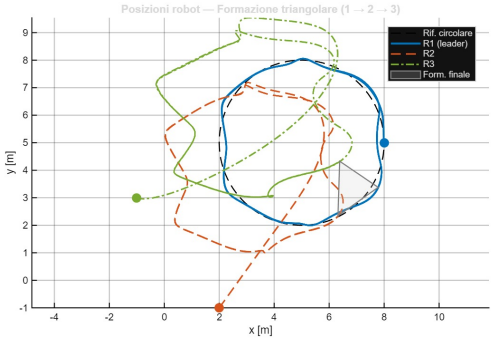


Figure 10: Agent trajectories subjected to combined sensor noise and persistent external disturbances.

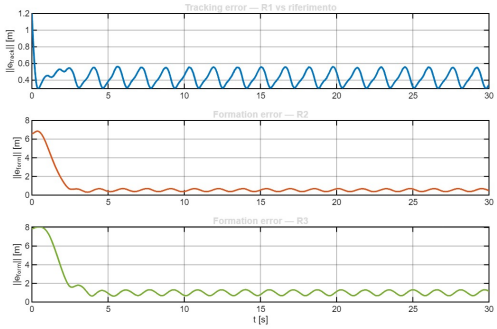


Figure 11: Tracking and formation errors under combined sensor noise and disturbances.

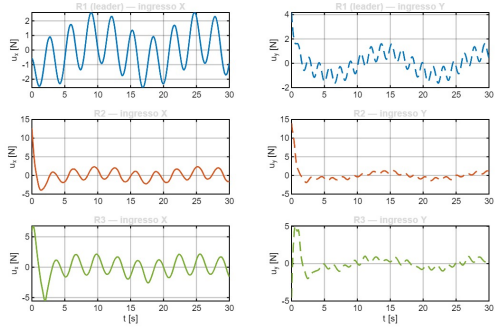


Figure 12: Control input force profiles under combined sensor noise and disturbances.

A comparative analysis of the final tracking errors reveals the structural inadequacy of standard state feedback control for real-world deployments. In the ideal scenario, the leader achieves

millimeter-level tracking precision (0.0000 m). However, the introduction of a constant external disturbance forces this steady-state error to jump to 0.0770 m, and a harmonic disturbance degrades it further to 0.2883 m, an error growth of nearly two orders of magnitude.

Crucially, because the tracking performance of the leader acts as the kinematic anchor for the entire network, any offset at node R_1 is inevitably transmitted downstream, causing the followers' formation errors to inflate beyond 0.5 m (R_3). The time-domain graphs clearly demonstrate that while state feedback is highly effective at absorbing initial geometric dislocations, its static structure is unable to reject persistent physical friction or fluctuating environmental forces. Consequently, deploying this architecture in unconstrained environments—naturally characterized by continuous sensor noise and unpredictable external drag—inevitably leads to unacceptable geometric distortion.

C. Conclusions and Transition to Loop Shaping

The inability of static state feedback to decouple low-frequency tracking from high-frequency noise rejection motivates a transition to the frequency domain. By adopting Loop Shaping, we replace the static gain with dynamic compensators $K(s)$ shaped to enforce high gain at low frequencies, guaranteeing disturbance rejection, while maintaining a steep magnitude roll-off at high frequencies to attenuate sensor noise.

V. Loop Shaping

A. Theoretical Framework for Distributed Loop Shaping

While the state-feedback approach relies on full-state information to compute the control action, the loop-shaping formulation achieves formation control by replacing the static state feedback with a dynamic output-feedback compensator $K(s)$. In this framework, the plant dynamics are described at the transfer-function level, and the control law is designed in the frequency domain.The target formation is specified entirely based on the system outputs, with desired output offsets defined as constants h_i. The fundamental objective is to drive the relative differences between the agents such that $y_i(t)-y_1(t) \to h_i-h_1$. To accomplish a moving formation, Agent 1 (the leader) tracks an external reference trajectory $r(t)$, meaning the desired global output for each agent becomes $y_i^*(t) = r(t) + h_i - h_1$. To formalize the distributed dynamic protocol, we first define the local output tracking error for each agent as:

$$\eta_i(t) = y_i(t) - r(t) - (h_i - h_1) \tag{50}$$

Relying exclusively on output information received from neighbors, the localized neighborhood formation error is computed as:

$$\epsilon_i(t) = \sum_{j=1}^{N} \mathcal{A}_{ij}(\eta_j - \eta_i) \tag{51}$$

To force the leader to track the reference, an additional local pinning control term is applied. This yields the pinned error signal:

$$\epsilon_i^p(t) = \epsilon_i(t) - \gamma\pi_i\eta_i(t) \tag{52}$$

where $\pi_1 = 1$ for the leader and $\pi_i = 0$ for all follower agents $i \neq 1$. The distributed loop-shaping control law is then synthesized to generate the virtual input for each agent as $u_i(s) = K(s)\epsilon_i^p(s)$. By aggregating the local tracking errors into a global vector $\eta = \text{col}(\eta_1, \ldots, \eta_N)$, the stacked control law for the entire multi-agent system can be expressed compactly as $u(s) = -K(s)\mathcal{L}_p\eta(s)$, where $\mathcal{L}_p = \mathcal{L} + \gamma\Pi_1$ represents the pinned Laplacian matrix. Provided that the communication graph contains a spanning tree and $\gamma > 0$, $\mathcal{L}_p$ is positive definite and possesses strictly positive eigenvalues $\rho_i > 0$. This matrix structure allows the multi-agent system to be decoupled into independent modal components by diagonalizing the pinned Laplacian. Consequently, the stability, disturbance rejection, and tracking performance of the network are entirely governed by the pinned modal sensitivity functions:

$$S_i^{pin}(s) = \frac{1}{1 + \rho_i G_p(s)K(s)} \tag{53}$$

The design challenge becomes a problem of simultaneous stabilization: we must find a single compensator $K(s)$ capable of stabilizing all modal loop functions $\rho_i G_p(s)K(s)$ concurrently. Internal stability is achieved if each modal loop satisfies the Nyquist criterion. To maintain the rigid geometric shape, the system must achieve zero steady-state formation error when tracking the constant output offsets. This requirement is met if and only if $S_i^{pin}(0) = 0$, which equivalently demands an infinite DC gain for the modal loop functions, expressed as $\lim_{s\to 0} |\rho_i G_p(s)K(s)| = \infty$. As established by control theory, this infinite

DC gain can be provided either by integrators inherent to the plant or by an explicit integral action within $K(s)$ itself. Following the MIMO input-output feedback linearization applied earlier, the individual agents are represented by an identical decoupled double-integrator plant, $G_p(s) = \frac{1}{s^2}$. While the plant $G_p(s)$ inherently possesses two poles at the origin, a Proportional-Integral-Derivative (PID) controller is actively chosen as the primary dynamic compensator $K(s)$ to guarantee structural robustness against external environmental conditions. The derivative action of the PID provides the necessary phase lead to stabilize the double integrators, ensuring internal stability. Simultaneously, the integral action is specifically incorporated to ensure that the modal loop functions forcefully attenuate any steady-state tracking offset induced by unmeasured, low-frequency constant output disturbances $d(s)$, such as wind or sloping terrain. In this distributed architecture, the integral term is practically anchored to the leader agent; by actively eliminating the leader's absolute tracking error, the entire multi-agent system is forced to remain strictly on the desired global trajectory while preserving the required geometric shape via the pinned formation loop.

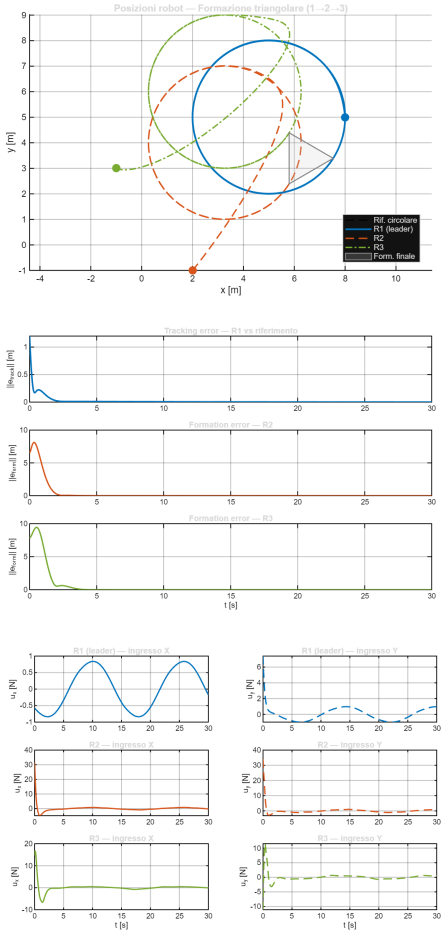


Figure 13: Nominal scenario results:
(a) Trajectories;
(b) Tracking and formation errors;
(c) Control efforts.

B. Experiments and Results

In order to validate the loop-shaping protocol, we implemented the dynamic compensator $K(s)$ designed in the frequency domain. Several simulations were conducted to demonstrate convergence towards the desired moving formation $y_i^*(t) = r(t) + h_i - h_1$ [**?**].

The system was further tested under actuator saturation constraints [**?**] and additive output disturbances, showing robust behavior in maintaining the formation despite these practical limitations.

C. Conclusions

The approach based on Loop Shaping proved extremely effective in replacing static state feedback with a dynamic compensator $K(s)$ based on output measurements [**?**]. Stability depends on the pinned modal sensitivity functions $S_i^p(s) = \frac{1}{1+\rho_i G_p(s) K(s)}$ [**?**]. The design required a trade-off: high gain at low frequencies for disturbance rejection and offset tracking, balanced against bandwidth limitations to prevent actuator saturation [**?**].

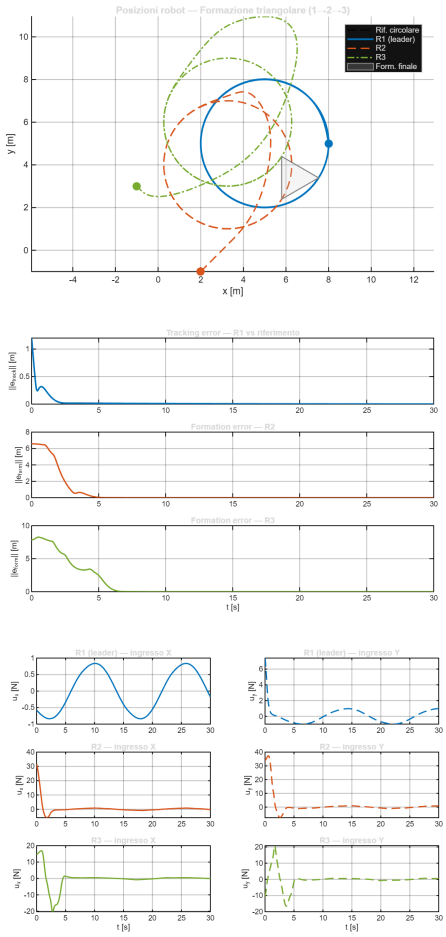


Figure 14: Actuator bounds scenario results:
(a) Trajectories;
(b) Tracking and formation errors;
(c) Control efforts.

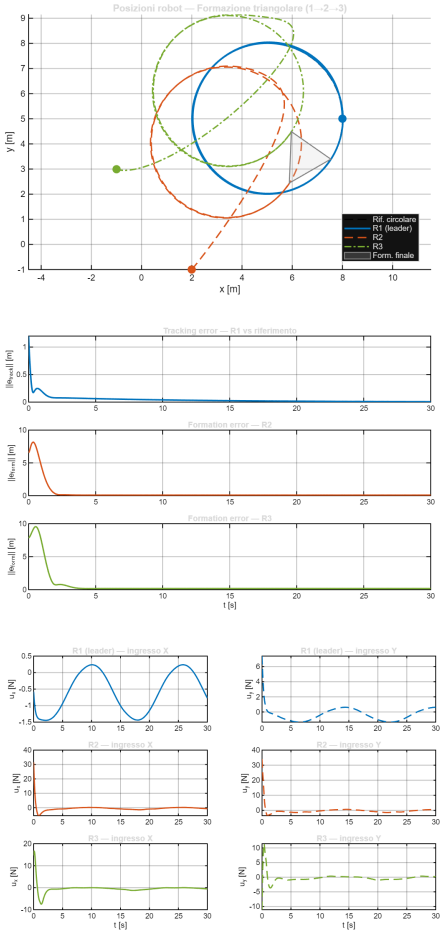


Figure 15: External constant output disturbances scenario results:
(a) Trajectories;
(b) Tracking and formation errors;
(c) Control efforts.

VI. Conclusion

In this project, a two-level control architecture was successfully developed to achieve a moving triangular formation for a team of heterogeneous planar mobile robots. First, local MIMO input-output feedback linearization effectively canceled the agents' specific nonlinearities, yielding a standardized double-integrator model that facilitated the design of a unified distributed controller.

While a State Feedback protocol proved effective under nominal conditions where higher graph connectivity optimized transient maneuvers robustness analyses exposed its severe vulnerability to persistent external disturbances and sensor noise. To overcome these structural limitations, a distributed loop shaping output feedback protocol was implemented. By incorporating a dynamic PID compensator with an integral action anchored at the leader node, the networked system successfully rejected low-frequency unmeasured disturbances and maintained strict geometric rigidity, even under actuator saturation constraints.Ultimately, the integration of local feedback linearization and dynamic frequency-domain compensation provides a highly robust and scalable framework for the coordination of cyber-physical systems in non-ideal environments.